Moran process

Three individuals, one die, one survivor

Set up
Work out the fitnesses by hand.

Roll
One birth, one death, every step.

Pool
Count how often the hawk fixes.

Pairs playing:

Starting state: one hawk, two doves

	Hawk	Dove
Hawk	0	3
Dove	1	2

Four resources to share. Two hawks get ____ each; a hawk meeting a dove takes ____ to the dove's ____; two doves split them ____ each.



The number of hawks. Along the top a hawk is born; along the bottom one dies. The two ends are absorbing, so the process stops only once one type is gone.

How fit is each type?

A type's fitness counts what it scores against everyone *else* in the population, so it never plays itself.

State	f(hawk)	f(dove)
1 hawk, 2 doves		
2 hawks, 1 dove		

Which type is fitter when hawks are rare? And when they are common?

Birth is proportional to fitness. Death is uniform, so it is proportional to _____.

State	Type	Chosen for birth	Chosen for death
1 hawk, 2 doves	hawk		
1 hawk, 2 doves	dove		
2 hawks, 1 dove	hawk		
2 hawks, 1 dove	dove		

These are the numbers the die has to reproduce. Every denominator divides twelve, which is why one D12 is enough.

Twelve faces split evenly into halves, thirds and quarters. A hawk is chosen when the roll lands in:

State	Hawk born on	Hawk dies on
1 hawk, 2 doves		
2 hawks, 1 dove		

Any higher roll chooses a dove.

In pairs, with one D12, starting from one hawk and two doves. Roll once for the birth and once for the death, and read each type off the table.

State	Hawk born on	Hawk dies on
1 hawk	**1 to 6**	**1 to 4**
2 hawks	**1 to 9**	**1 to 8**

Keep count of the hawks. When the count reaches 0 or 3, make a tally mark for how that run ended and start again from one hawk.

For example, with one hawk a birth roll of 3 and a death roll of 9 give a new hawk and one fewer dove, so we move to two hawks.

Each run

1. Start with **1 hawk**, 2 doves.
2. Roll for the **birth**, then for the **death**.
3. Write down the rolls and the new number of hawks.
4. Repeat until there are **0 or 3** hawks.
5. Make a tally mark, then start a new run.

Hawks	Hawk born on	Hawk dies on
1	**1 to 6**	**1 to 4**
2	**1 to 9**	**1 to 8**

Any higher roll is a dove.

What your paper should look like

Hawks	Birth	Death	Hawks next
1	3 (hawk)	9 (dove)	2
2	11 (dove)	2 (hawk)	1
1	5 (hawk)	2 (hawk)	1
1	8 (dove)	1 (hawk)	**0**

Run over: all doves.

All hawks: |||

All doves: ||||

When we stop, each pair reports two numbers: how many runs ended all hawks, and how many ended all doves.

The number of hawks changes only when the type born differs from the type that dies.

$p_{10} =$ $p_{12} =$ $p_{21} =$ $p_{23} =$

Go back and write these on the chain. What happens with the remaining probability?

Runs that ended with all hawks: ____ out of ____, so a fixation probability of about ____.

The theory says $x_1 = \dfrac{1}{1 + \gamma_1 + \gamma_1\gamma_2}$ with $\gamma_i =$ ____, giving ____.

Close enough? ____ How many runs would we need to be sure? ____

What we just did in the room is **Question 1** on the Moran Process page, with a population of four instead of three.

(a)	The two random choices made at each step.	[3]
(b)	Fitnesses with two hawks and two doves.	[6]
(c)	The probability the number of hawks increases by one.	[7]
(d)	Define fixation, and say why there are exactly two absorbing states.	[5]
(e)	The probability it decreases, the ratio γ, and which way it drifts.	[4]

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Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Three individuals, two types	**11.2.1**	Moran Process: Definition: Moran Process
Fitness from the payoff matrix	**11.2.1**	Moran Process: Definition: Moran Process
Counting how often hawks took over	**11.2.2**	Moran Process: Definition: The Fixation Probability
Selection strength	**11.2.6 to 11.2.8**	Moran Process: Neutral Drift, Strong Selection, Weak Selection
Examinable	**11.2.3**	Theorem: The fixation probabilities in populations of two types, and proof

vknight.org/gtb/chapters/moran-process

Quick quiz before you go: vknight.org/gt/topics/moran-process.html

Before the next class: read the Moran Process chapter as far as the fixation probabilities, and attempt Question 1.